

Week 1 - Introduction to AI Engineering

Concepts:

Overview of AI Engineering, AI Engineering vs AI Research, Career Paths in AI Engineering

Project:

Create a personal AI Engineering portfolio

Outcome:

Understand the basics of AI Engineering and its career paths

Week 2 - Mathematics for AI

Concepts:

Linear Algebra, Calculus, Probability and Statistics

Project:

Solve a linear algebra problem using Python

Outcome:

Understand the mathematical concepts required for AI Engineering

Week 3 - Programming Fundamentals

Concepts:

Python Programming, Data Structures and Algorithms

Project:

Implement a data structure using Python

Outcome:

Develop programming skills required for AI Engineering

Week 4 - Introduction to Deep Learning

Concepts:

Neural Networks, Activation Functions, Backpropagation

Project:

Implement a simple neural network using Python

Outcome:

Understand the basics of deep learning

Week 5 - Deep Learning Frameworks

Concepts:

TensorFlow, Keras, PyTorch

Project:

Implement a deep learning model using TensorFlow

Outcome:

Understand popular deep learning frameworks

Week 6 - Computer Vision

Concepts:

Image Processing, Object Detection, Image Segmentation

Project:

Implement a computer vision task using OpenCV

Outcome:

Understand the basics of computer vision

Week 7 - Natural Language Processing

Concepts:

Text Processing, Sentiment Analysis, Named Entity Recognition

Project:

Implement a natural language processing task using NLTK

Outcome:

Understand the basics of natural language processing

Week 8 - Reinforcement Learning

Concepts:

Markov Decision Processes, Q-Learning, Policy Gradient Methods

Project:

Implement a reinforcement learning task using Gym

Outcome:

Understand the basics of reinforcement learning

Week 9 - Deep Learning Applications

Concepts:

Image Classification, Object Detection, Speech Recognition

Project:

Implement a deep learning application using TensorFlow

Outcome:

Understand the applications of deep learning

Week 10 - AI Ethics and Bias

Concepts:

AI Ethics, Bias in AI, Fairness in AI

Project:

Analyze the ethics of a deep learning model

Outcome:

Understand the ethics and bias in AI

Week 11 - AI Engineering Project

Concepts:

Project Planning, Project Execution, Project Monitoring

Project:

Implement an AI engineering project

Outcome:

Apply AI engineering skills to a real-world project

Week 12 - Career Development

Concepts:

Resume Building, Interview Preparation, Networking

Project:

Update a resume and prepare for an AI engineering interview

Outcome:

Prepare for a career in AI engineering
